

Developing a Graph-based Visualization of Elective Courses to Support Course Selection in Higher Education

Marie Trippel ¹ and René Röpke ²

Abstract: In an open curriculum in higher education, students face the challenge of navigating a broad course catalog independently. Prerequisites and recommended courses can make this even more complex, as they are not always easy to follow. Providing students with clear information is essential to make good choices in study planning. Effective visualization could simplify this process for students by providing structured, understandable, and relevant information. This paper presents a novel tool that helps students choose their elective courses using a graph-based visualization of course relationships, emphasizing prerequisites and recommendations. A first user evaluation indicates a strong interest in a course-planning tool for electives and the use of the graph-based visualization was well-liked. Thereby, this research highlights students' demand for a planning tool and demonstrates the potential of graph visualizations as a valuable resource for future development in this area.

Keywords: Course Planner, Dependency Visualization, Course Prerequisites

1 Motivation and Related Work

An open curriculum in higher education is flexible and allows personalization based on interests, giving students greater autonomy in shaping their academic journeys. However, this increased flexibility also introduces complexities in course selection, often leaving students in need of additional support to navigate their options effectively and consequently have successful studies. Often, students only have limited knowledge about possible choices as well as regulations and restrictions. This can lead to a non-ideal study path if they do not know or miss important information, like course dependencies or semester availability.

To address these challenges, higher education institutions (HEI) provide advisory services and information materials. Traditionally, study advisors help students, offering tailored guidance. While being invaluable, especially in special circumstances, this support is resource-intensive and time-consuming. In recent years, HEI started to develop digital study planning and assistance tools, empowering students to take charge of their study planning and align it with individual requirements and objectives [Na24; Ru24; TP18; Un24a; Un24b]. Within these tools, relevant curriculum information needed for study planning is presented and the planning process can be supported through visual depiction of semesters and course order [HHH21; Ju23a; Ju23b].

¹ RWTH Aachen University, Aachen, Germany,
marie.trippel@rwth-aachen.de,  <https://orcid.org/0000-0003-4961-986X>

² TU Wien, Learning Technologies and eDidactics, Vienna, Austria,
rene.roepke@tuwien.ac.at,  <https://orcid.org/0000-0003-0250-8521>

Visualization is a powerful tool for presenting complex information clearly and understandable [Sh96]. It presents as a promising approach to make complex curricula, including course relations and dependencies, more accessible and easily comprehensible. While there are various types of visualizations for types of data [NJG21], for this work, graph-based visualization is considered—a suitable approach to visualize interrelated entities (i.e. courses and their relations), as seen in social networks [MR20; Ne13].

This work introduces an interactive graph-based visualization tool to assist students in assessing course options and their relations, and selecting suitable courses, thus enabling informed decision-making. As for now, the tool relies solely on data from available module manuals/handbooks, with means to integrate it into the tool landscape of HEI. While it does not yet consider how courses complement each other beyond listed prerequisites, the tool provides a work-in-progress prototype for further investigation and extension.

2 System Design and Implementation

The primary objective of the graph-based tool is to visualize courses, their relations and dependencies, their allocation in different subject areas and additional course information. Courses and subject areas can both be depicted as two types of nodes in graph: Circular nodes for courses and square nodes for subject areas. Relations are depicted as directed edges and two types of course-to-course relations are considered: “recommended”, if A is recommended for B, and “required”, if A is a prerequisite for B, where A and B are courses in the curriculum. Color is used to distinguish the directed edges: “required” edges are red, while “recommended” edges are blue. To visualize the curriculum structure, subject areas are connected to courses via non-directed containment edges (colored in grey). For displaying additional course information, further visual cues are used: node size indicates a course’s credits, while a green ring signifies if a course has been passed.

Requirement Analysis A requirement analysis was conducted using semi-structured interviews³ to ensure tool meets user needs. Four key stakeholders were interviewed: the head of the Study Center, the Study Program Coordinator, Student Member of the Examination Committee, and an member of the Academic Advisory Service. Their input provided functional requirements for the tool (see Table 1).

Implementation The tool was implemented as an Angular-based web application [Go25] using d3.js [Bo25] for the graph visualization due to its low-level DOM (Document Object Model) control. The component-based design enables the resulting tool to be easily integrated in existing applications, e.g., study planning tools.

Implemented features include filtering by subject areas, credits, semester cycle, course language, and course status, as well as preset filters for quick setup (FR6). Users can hover to highlight linked courses (FR4), click to view course information (FR3, FR5), and

³ Research data is available under <https://doi.org/10.17605/OSF.IO/ZTQ2U>

ID	Description
FR1	Courses should be represented as nodes in the graph.
FR2	Prerequisite relations should be represented as a directed edge.
FR3	Nodes should display course name, course area, and credits.
FR4	Users should be able to interact with the graph (e.g., zoom, pan, click on nodes).
FR5	When hovering over or clicking on a node, more course information should appear.
FR6	Users should be able to filter courses based on criteria such as department, level, or semester.
FR7	Users should be able to search for specific courses by name or code.
FR8	Users can set each course to passed, taken, or planned to be taken.
FR9	Users can manage multiple individual pathways.
FR10	When hovering over a course node, all connected nodes should be highlighted.
FR11	The system can offer preset views (e.g., by departments, specializations).
FR12	The system should warn users if a course has requisites that are not in the module catalog.
FR13	Users can create and add new course nodes to the graph.
FR14	Users can rate courses, thereby providing recommendations for other users.
FR15	The system can recommend similar subjects based on course selections.
FR16	The system should warn users if their course selection is too specialized or too broad.

Tab. 1: List of determined functional requirements

toggle graph options such as shortened titles or visualizing prerequisite paths. The tool is standalone and not yet integrated in any study planning tool and stores user data only locally, which is why a export/import functionality is provided for returning users (FR9). Warnings appear when a student selects too few or too many courses in one area (FR16).

3 Evaluation

A user study was conducted to evaluate the work-in-progress prototype and collect first feedback on its features and design. The study involved n=15 participants, all Computer Science (CS) students, enrolled in Bachelor or Master study programs and with prior experience in choosing electives as part of their curriculum. The study was designed as a scenario-based evaluation using the think-aloud method, a short survey on user experience and a brief interview afterwards. The first scenario focused on choosing a course for the winter semester with four credits, to get the students familiar with the interface and functionalities. The second scenario consisted of looking for a specific CS elective course that has prerequisites and seeing whether this course is feasible. Here students had to navigate through the interface and understand a course path with two prerequisites.

3.1 Qualitative Insights from the Evaluation and Interviews

This section presents the qualitative results of the scenario-based evaluation and the interview. All participants reported a positive experience with the tool, with most participants expressing a desire to use it for their own study planning. The prototype helped to think more strategically

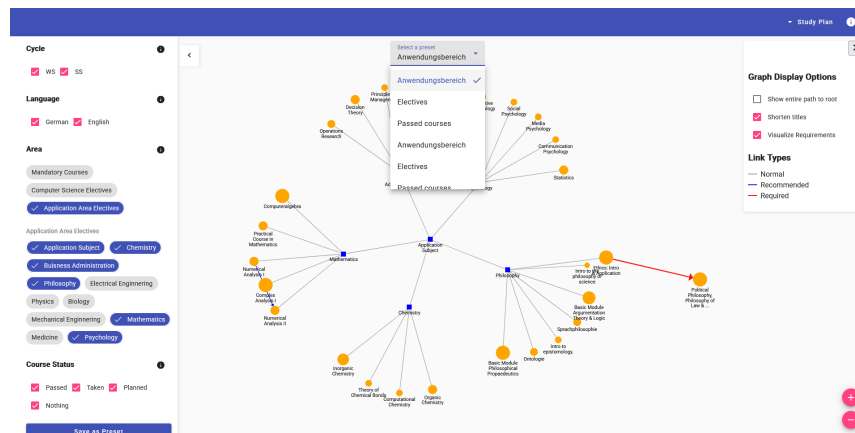


Fig. 1: At the core of the application is the graph structure visualizing a curriculum with courses as nodes (FR1, FR3). A filter menu is on the left side (FR6), integrated in a sidebar. In the top center, the preset drop-down (FR11) is located. On the top right is a box with graph display options and a legend of the different link types (edges). A course recommendation is visualized by blue directed arrows (FR2). Required edges can be recognized by their red color as well as the path forming to further visualize the requirement (FR2).

about course selection, especially in understanding course relations. In the past, as most participants reported, they had chosen electives rather spontaneously.

The graph visualization of the curriculum was well-received, as it helped participants grasp the range of available options. Many participants appreciated the clear depiction of prerequisites through “recommended” and “required” links (directed, colored edges), and the filter functionality for narrowing down choices. They found it less overwhelming than going through a traditional list, like in their campus management system (CMS) or the module manual. Participants also appreciated tracking course status for long-term planning.

The interface initially appeared overwhelming to some, mainly because of the large graph structure. However, they quickly adapted by using the preset and filter options to get an overview. Interestingly, some participants relied solely on the preset options, while others exclusively used the filter menu. While the preset options sped up the process, some participants found their connection to the filters unclear, calling for clarification in its design.

Participants expressed interest in more detailed course information such as expected workload, information on the instructors, and class schedule. They recommended adding direct links to registration pages, as well as past evaluations or data on average grades. The most requested, yet missing feature was a search function (FR7). Others included side-by-side course comparisons and a course recommendation system providing suitable, personalized course choices.

3.2 Survey Results on User Experience

Next, we present results from the survey portion of our study, evaluating the participants' user experience when testing the tool and completing the scenario-based evaluation. For this, the User Experience (UEQ) was used due to ease of use, wide applicability and reliability [LHS08]. Overall, scores were good across all six dimensions, indicating a generally positive user experience. The dimension "Stimulation" ($M = 1.900$, $SD = 0.667$) and Attractiveness ($M = 1.811$, $SD = 0.842$) scored highest, showing that participants found the tool engaging and pleasant to use. "Efficiency" ($M = 1.617$, $SD = 0.812$), "Dependability" ($M = 1.650$, $SD = 0.766$) and "Perspicuity" ($M = 1.367$, $SD = 0.855$) were rated positively but suggest room for improvement in clarity and task flow. These results are consistent with qualitative feedback on unclear UI (User Interface) elements and the similarity of visual indicators. Results for the "Novelty" dimension ($M = 1.733$, $SD = 0.665$) indicate that participants perceive the tool as innovative and creative. The tool scored overall higher in Hedonic Quality than in Pragmatic Quality ($1.82 > 1.54$), indicating that while participants found it enjoyable and appealing, its usability and functionality can still be improved to better support students' needs.

4 Discussion

Overall, we were able to design and implement a graph-based visualization tool, fulfilling most of the collected requirements. As this prototype presents is work-in-progress, our first goal was to collect feedback by students and assess user experience. The data used in the tool was solely based on module manuals and did not integrate data of a CMS.

Both, the qualitative insights from the scenario-based evaluation and the interview, as well as the quantitative results on the participants' user experience suggest that students are eager for a tool to assist them in choosing elective courses. Especially during the interviews, participants voiced interest in using it for their own study planning in order to compare electives and to make informed choices.

While some participants had initial difficulties grasping the structure of the very large curriculum graph, provided filters and preset options helped navigating and getting oriented quickly. The graph structure itself was easy to understand and the provided information was perceived as helpful, yet, it could be extended in the future. Participants asked for information that could be easily provided from CMS (e.g., instructor information or class schedules), but they also showed interest in data-driven insights (e.g., grade statistics from past semesters). This calls for its integration in an institution's tool landscape as well as the extension with learning analytics. Through learning analytics, data-driven insights on courses could be provided to students when comparing courses.

Limitations This ongoing work has several limitations which need to be addressed. First, the prototype was evaluated exclusively within the CS faculty at RWTH Aachen University,

where the Bachelor study program has rather many restrictions in early semesters and then moves towards almost no restrictions, but only recommendations linking different courses to each other. It was therefore also only tested with CS students, who may have a higher affinity for new digital tools and learn about graph structures as part of their curriculum. This may have biased the results and calls for the evaluation with students from other study programs. Additionally, as already mentioned, this work focused solely on visualizing course relations based on the module manual and did not incorporate any recommendation systems, or look deeper into which courses complement each other based on their actual content. This work is, thus, limited to the quality and availability of data provided in module manuals. Further, informal knowledge of course orders due to course content and difficulty is also not considered. Lastly, as this work does not examine social and educational impact yet, a long-term evaluation would be interesting.

Future Work With this work presenting a prototype, future work calls for user-centered evaluation and systematic integration in HEI. The feedback so far should be considered for implementation, along with missing requirements (e.g., FR7, FR14). The tool should also be tested with students from other study programs to see how results differ when there are more (or less) course prerequisites, and to assess the user experience of students who may be less tech-savvy than CS students. Difficulties with the graph visualization could arise which may require adjustments in the tool design or additional support in learning to use the tool. It should also be considered to explore alternative visualizations, e.g., representing courses within subject areas using bubble charts. Furthermore, the combination of graph-visualization with other visualizations could be explored to provide various ways to interact with a curriculum. Lastly, when developing the prototype further, its integration in study planning tools or CMS should be considered to allow for smooth transitions and ease-of-use for students.

5 Conclusion

This paper addressed the challenge of students when selecting elective courses in multi-semester planning, with a focus on course relations and dependencies. To this, we introduced a graph-based visualization tool to support students in exploring the curriculum and in making informed course choices. The development process included a requirement analysis with stakeholders and the design and implementation as a web application. The resulting prototype was evaluated in a user study with 15 participants to receive feedback and gain insights on its design and the user experience. Results indicate a strong potential and interest for graph-based visualizations for students' course selection. Next steps include improving and further testing the prototype as well as integrating it existing tools for study planning. With it being integrated in tools available to students, further investigations of its benefits to study planning and guidance can be conducted providing valuable insights into students' perspectives.

References

- [Bo25] Bostock, Mike and Observable, Inc.: D3 by Observable – The JavaScript library for bespoke data visualization, 2025, <https://d3js.org/>, accessed: 04/06/2025.
- [Go25] Google: Angular, 2025, <https://angular.dev/>, accessed: 04/06/2025.
- [HHH21] Hirmer, T.; Heyne, N.; Henrich, A.: Die kompetenzorientierte Studienplanung - Entwicklung eines Tools zur Unterstützung von (Lehramts-)Studierenden. In: DELFI 2021. Gesellschaft für Informatik e.V., Bonn, pp. 121–126, 2021.
- [Ju23a] Judel, S. et al.: KI-gestützte Studienverlaufsplanung und -analyse mit AIStudyBuddy. In: Workshops der 21. Fachtagung Bildungstechnologien (DELFI). Gesellschaft für Informatik e.V., Bonn, pp. 193–196, 2023, DOI: 10.18420/WSDELFI2023-55.
- [Ju23b] Judel, S. et al.: Supporting Individualized Study Paths Using an Interactive Study Planning Tool. In: 21. Fachtagung Bildungstechnologien (DELFI). Gesellschaft für Informatik e.V., Bonn, pp. 225–230, 2023, DOI: 10.18420/DELFI2023-36.
- [LHS08] Laugwitz, B.; Held, T.; Schrepp, M.: Construction and Evaluation of a User Experience Questionnaire. In: Symposium of the Austrian HCI and Usability Engineering Group. Springer, pp. 63–76, 2008, DOI: 10.1007/978-3-540-89350-9_6.
- [MR20] Majeed, A.; Rauf, I.: Graph Theory: A Comprehensive Survey about Graph Theory Applications in Computer Science and Social Networks. *Inventions* 5 (1), p. 10, 2020, DOI: 10.3390/inventions5010010.
- [Na24] National University of Singapore: NUSMODS, <https://nusmods.com/timetable>, 2024, <https://nusmods.com/timetable>, accessed: 04/13/2024.
- [Ne13] Nettleton, D. F.: Data mining of social networks represented as graphs. *Computer Science Review* 7, pp. 1–34, 2013, DOI: 10.1016/j.cosrev.2012.12.001.
- [NJG21] Nguyen, V. T.; Jung, K.; Gupta, V.: Examining data visualization pitfalls in scientific publications. *Visual Computing for Industry, Biomedicine, and Art* 4 (1), p. 27, 2021, DOI: 10.1186/s42492-021-00092-y.
- [Ru24] Rutgers University: Rutgers Course Schedule Planner, 2024, <https://sims.rutgers.edu/csp/>, accessed: 08/30/2024.
- [Sh96] Shneiderman, B.: The eyes have it: a task by data type taxonomy for information visualizations. In: *Proceedings 1996 IEEE Symposium on Visual Languages*. IEEE Comput. Soc. Press, Boulder, CO, USA, pp. 336–343, 1996, DOI: 10.1109/VL.1996.545307.
- [TP18] Tomy, S.; Pardede, E.: Course Map: A Career-Driven Course Planning Tool. In (Gervasi, O. et al., eds.): *Computational Science and Its Applications – ICCSA 2018*. Vol. 10961, Springer International Publishing, Cham, pp. 185–198, 2018, DOI: 10.1007/978-3-319-95165-2_13.
- [Un24a] Universeiteit Utrecht: CoursePlanner Utrecht University, <https://cursusplanner.uu.nl>, 2024, <https://cursusplanner.uu.nl>, accessed: 08/30/2024.
- [Un24b] University of Helsinki: Sisu Instructions: Study Plan, 2024, <https://studies.helsinki.fi/instructions/article/sisu-instructions-study-plan>, accessed: 09/06/2024.